CLASS 9 · CBSE · GANITA MANJARI PART 1 · DPP SHEET

Chapter 5: I'm Up and Down, and Round and Round

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. A chord is 12 cm from the centre of a circle of radius 13 cm. Find the length of the chord.
2. A chord 16 cm long is 15 cm from the centre. Find the radius.
3. A circle has diameter 20 cm. How far from the centre is a chord of length 12 cm?
4. An arc subtends 84° at the centre. What angle does it subtend at a point on the remaining part of the circle?
5. AB is a diameter and C is a point on the circle with $\angle ABC = 28^\circ$. Find $\angle BAC$.
6. In a cyclic quadrilateral $PQRS$, $\angle P = 68^\circ$ and $\angle Q = 105^\circ$. Find $\angle R$ and $\angle S$.
7. In a cyclic quadrilateral $ABCD$, $\angle A = (4x - 10)^\circ$ and $\angle C = (x + 40)^\circ$. Find x.
8. Two parallel chords of lengths 14 cm and 48 cm lie on the same side of the centre of a circle of radius 25 cm. Find the distance between them.
9. A chord subtends 60° at the centre of a circle of radius 7 cm. Find its length.
10. A regular hexagon is inscribed in a circle of radius 8 cm. Find the length of each side.

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Answer Key

1. $2\sqrt{169-144} = 10$ cm
2. $\sqrt{8^2+15^2} = 17$ cm
3. $\sqrt{10^2-6^2} = 8$ cm
4. 42°
5. $90^\circ - 28^\circ = 62^\circ$
6. $\angle R = 112^\circ$ and $\angle S = 75^\circ$
7. $5x + 30 = 180$, so $x = 30$
8. The distances from the centre are 24 cm and 7 cm, so the chords are 17 cm apart.
9. 7 cm (the triangle is equilateral)
10. 8 cm (each side subtends 60° at the centre)

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